

Network Security 1 Rubric 4

This assignment is about operating-system/user access control on the Mac Book that hosts my home lab.

User management is a critical part of maintaining a secure system. Ineffective user and privilege management often leads to a system being compromised. Therefore, it is important that individuals in cybersecurity understand how to protect their servers through simple and effective user account management techniques.

For my lab, the cleanest implementation is to use **Ubuntu** as the protected computer and demonstrate:

1. **ACL configuration:** Linux file-system ACL using `setfacl`
2. **Access control model: DAC (Discretionary Access Control)**
3. **User access level:** A standard user versus an administrator/root user

Recommended Home Lab Implementation

Use the Ubuntu protected system:

Ubuntu Protected Computer

```
|  
+-- Administrator account  
|  
+-- Analyst user
```

|
+-- Protected project files

Example scenario

Create two Ubuntu users:

- `adminuser` → administrator-level user

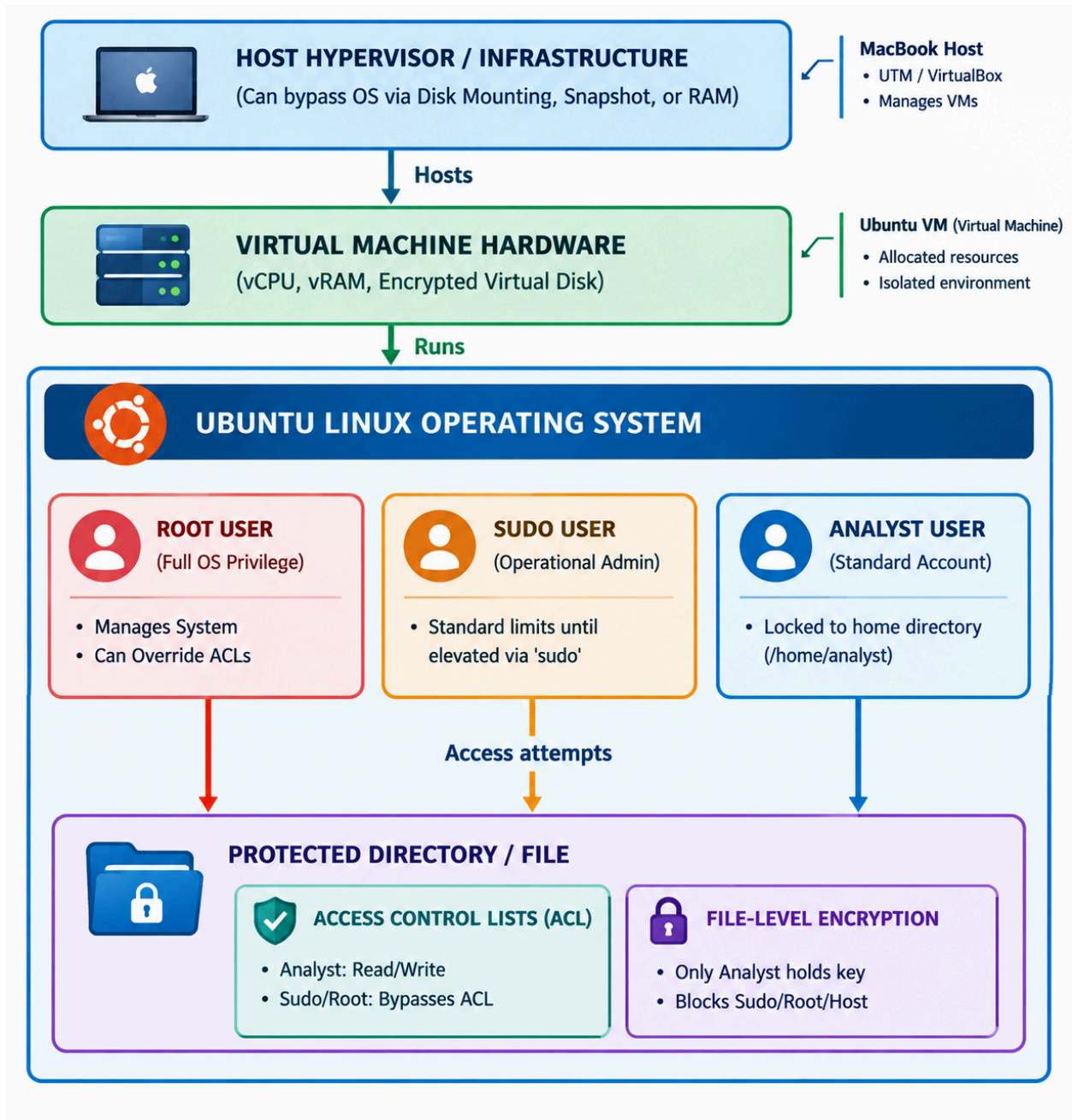
Then create a protected directory:

`/home/labdata`

The administrator owns the directory, while the analyst receives **read-only access** through a Linux ACL.

That gives you a very clear demonstration:

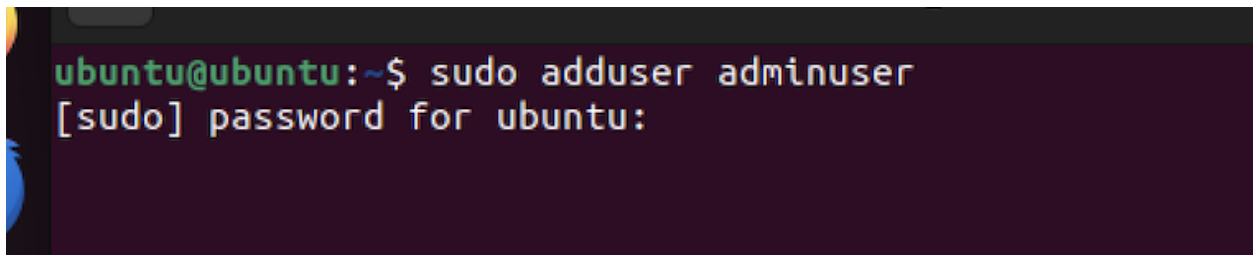
Requirement	Your Home Lab Implementation
ACL configuration	Linux file-system ACL using <code>setfacl</code>
Access control model	DAC
Protected computer	Ubuntu
Protected resource	<code>/home/labdata</code>
Administrator	<code>adminuser</code>
Restricted user	<code>analyst</code>



1. Create the Users

On Ubuntu, create the administrator and analyst accounts.

```
sudo adduser adminuser
```

A terminal window with a dark purple background. The prompt 'ubuntu@ubuntu:~\$' is followed by the command 'sudo adduser adminuser'. Below this, the prompt '[sudo] password for ubuntu:' is displayed, with the rest of the terminal area obscured by a dark rectangle.

```
ubuntu@ubuntu:~$ sudo adduser adminuser
[sudo] password for ubuntu:
```

Then:

```
sudo adduser analyst
```

You can make `adminuser` an administrator:

```
sudo usermod -aG sudo adminuser
```

The `analyst` account remains a **standard user**.

You can verify the groups:

```
groups adminuser
```

and:

```
groups analyst
```

The important difference is that `adminuser` has administrative privileges through the `sudo` group, while the `analyst` does not.

2. Create a Protected Resource

Create a directory for sensitive lab information:

```
sudo mkdir /home/labdata
```

Create a test file:

```
sudo touch /home/labdata/security_report.txt
```

Put some information into it:

```
echo "Cybersecurity Home Lab Security Report" | sudo tee  
/home/labdata/security_report.txt
```

Make `adminuser` the owner:

```
sudo chown -R adminuser:adminuser /home/labdata
```

3. Configure the ACL

Now give the `analyst` user **read-only access** to the directory and its contents:

```
sudo setfacl -m u:analyst:rx /home/labdata
```

And give the analyst read access to the report:

```
sudo setfacl -m u:analyst:r /home/labdata/security_report.txt
```

You can inspect the ACL with:

```
getfacl /home/labdata
```

and:

```
getfacl /home/labdata/security_report.txt
```

You should see an entry for:

```
user:analyst:r--
```

or equivalent ACL information.

4. Test the User's Access

Switch to the analyst account:

```
su - analyst
```

Try to read the report:

```
cat /home/labdata/security_report.txt
```

The analyst should be able to **read** the file.

Now try to modify it:

```
echo "Unauthorized modification" >> /home/labdata/security_report.txt
```

The operation should be denied because the analyst has read-only access.

This is excellent evidence for your assignment because you can demonstrate:

Allowed:

Analyst → Read security_report.txt → ALLOWED

Denied:

Analyst → Modify security_report.txt → DENIED

5. Access Control Model: DAC

Your access control model is **Discretionary Access Control (DAC)**.

Linux file permissions and ACLs are a good example of DAC because the **owner of a resource can control who has access to it**.

In your lab:

adminuser

|

| owns

v

/home/labdata

|

+---- security_report.txt

|

+---- analyst: READ ONLY

The administrator/owner controls the permissions granted to the analyst.

Why DAC?

DAC is appropriate for your project because it allows the owner of a computer resource to decide which users can access that resource and what operations they can perform.

6. User Access Level

Your required user access level can be:

Standard User — **analyst**

The **analyst** account represents a cybersecurity analyst who needs access to information but should not have unrestricted administrative privileges.

The analyst can:

- Log into Ubuntu
- Read authorized security reports
- Analyze files
- Run normal user commands
- Access assigned lab resources

The analyst cannot:

- Modify protected reports
- Change system-wide settings
- Install software requiring administrator privileges
- Modify other users
- Perform unrestricted administrative operations

Administrator — **adminuser**

The administrator has elevated privileges and can:

- Modify system configuration
- Manage users
- Change permissions

- Configure ACLs
- Install software
- Perform administrative maintenance

This demonstrates **least privilege**: the analyst receives only the permissions needed for their role.

7. Your Assignment Report

You can use this directly in your report:

Access Control Implementation – Cybersecurity Home Lab

1. Introduction

The purpose of this implementation is to demonstrate access control measures within the cybersecurity home lab. The implementation focuses on controlling access to resources on a computer system rather than controlling network traffic.

Ubuntu is used as the protected computer in the home lab. Two user accounts are created to demonstrate different access levels: an administrator account and a standard cybersecurity analyst account. A Linux file-system Access Control List (ACL) is then configured to control the analyst's access to a protected security report.

The implementation demonstrates three required access control concepts:

1. One Access Control List (ACL) configuration.

2. One access control model: Discretionary Access Control (DAC).
3. One user access level: Standard User.

2. Home Lab Environment

The access control implementation is performed on the Ubuntu protected system used in the cybersecurity home lab.

Component	Configuration
Protected computer	Ubuntu Linux
Administrator account	adminuser
Standard user	analyst
Protected resource	<code>/home/labdata</code>
Protected file	<code>security_report.txt</code>
ACL technology	Linux POSIX ACL
Access control model	Discretionary Access Control (DAC)

The purpose of this configuration is to simulate how access to cybersecurity information can be restricted based on the user's role and required privileges.

3. User Accounts and Access Levels

Two user accounts are created on the Ubuntu system.

Administrator User

The `adminuser` account is assigned administrative privileges through the Linux `sudo` group. This user is responsible for system administration, user management, permission changes, and security configuration.

Standard Analyst User

The `analyst` account is configured as a standard user. The analyst does not receive unrestricted administrative privileges.

The analyst is permitted to access specific cybersecurity information required for analysis but is prevented from modifying protected information.

This separation demonstrates the principle of least privilege.

4. Access Control List Configuration

A protected directory named `/home/labdata` is created to store cybersecurity-related information.

A file named `security_report.txt` is placed inside the directory.

The directory is owned by the administrator:

```
sudo chown -R adminuser:adminuser /home/labdata
```

An ACL is then configured to provide the `analyst` user with read and directory traversal permissions:

```
sudo setfacl -m u:analyst:rx /home/labdata
```

The analyst is given read-only access to the security report:

```
sudo setfacl -m u:analyst:r /home/labdata/security_report.txt
```

The ACL can be verified using:

```
getfacl /home/labdata/security_report.txt
```

The ACL provides the following access:

User	Resource	Permission	Result
adminuser	security_report.txt	Read/Write	Allowed
analyst	security_report.txt	Read	Allowed
analyst	security_report.txt	Write	Denied

5. Access Control Model – DAC

The access control model used in this implementation is Discretionary Access Control (DAC).

DAC allows the owner of a resource to determine which users can access the resource and what permissions those users receive.

In this home lab, **adminuser** owns the protected directory and security report. An ACL is used to give the **analyst** account specific permissions without giving the analyst ownership of the resource.

This demonstrates how a system administrator can provide users with only the access required for their job.

6. User Access Level

The selected user access level is Standard User.

The **analyst** account represents a cybersecurity analyst who needs to review security information but does not require unrestricted control of the Ubuntu computer.

The analyst is allowed to:

- Log into the Ubuntu system.
- Access authorized security information.
- Read the security report.
- Perform normal user operations.

The analyst is not allowed to:

- Modify the protected security report.
- Change the ACL.
- Manage other users.
- Make unrestricted system-level changes.
- Perform administrative operations requiring root privileges.

7. Access Verification

The ACL configuration is tested by switching to the analyst account:

```
su - analyst
```

The analyst attempts to read the protected report:

```
cat /home/labdata/security_report.txt
```

The read operation is allowed because the ACL grants the analyst read permission.

The analyst then attempts to modify the report:

```
echo "Unauthorized modification" >> /home/labdata/security_report.txt
```

The modification should be denied because the analyst has read-only access.

This provides evidence that the ACL is enforcing the intended access restrictions.

8. Security Benefits

This implementation provides several security benefits.

First, it separates administrative privileges from normal user privileges. Second, it prevents a standard analyst account from modifying protected security information. Third, the ACL provides more specific control than simply giving the user full access to the directory.

The configuration also demonstrates the principle of least privilege, where users receive only the permissions necessary to perform their assigned responsibilities.

9. Conclusion

The access control implementation successfully demonstrates computer-based access control within the cybersecurity home lab.

A Linux file-system ACL was configured to give the **analyst** user read-only access to a protected security report. Discretionary Access Control (DAC) was used as the access control model, with the administrator controlling access to the protected resource. The analyst was assigned a standard user access level rather than unrestricted administrative privileges.

The implementation demonstrates how user accounts, access levels, file permissions, and ACLs can work together to protect sensitive information on a computer system.

Application ACL

Commands used in Kali Linux to add a new user, add privilege for the user and remove the user.

cd /etc

The command `cd /etc` [changes your current working directory in the terminal to the system configuration folder named /etc.](#)

How It Works

- **cd:** Stands for "change directory". It moves you from your current folder to a new folder.
- **/etc:** An absolute path pointing to the system-wide configuration directory.

What the /etc Folder Contains

- **Configuration files:** Text files that control how your operating system and installed programs behave.
- **System settings:** Network configurations, user account information, and service startup scripts.

The command `ls` will display all the files in this directory

```
(kali@kali)~$ cd /etc
cd: no such file or directory: etchome/kali
(kali@kali)~$ cd /etc
(kali@kali)~/etc$ ls
adduser.conf      cron.daily        ethertypes        hdparm.conf       ld.so.cache       mime.types         odbc.ini          pulse             screenrc          subuid            UPower
alternatives      cron.hourly       ettercap           host.conf          ld.so.conf         minicom            odbcinst.ini     python2.7         sddm.conf.d      subuid-          usb_modeswitch.conf
apache2           cron.monthly      ferretbuster       hostname           ld.so.conf.d       miredo             OpenCL            python3.13        searchplot_rc    subversinn       usb_modeswitch.d
apparmor          cron.weekly       firefox-esr        hosts             libao.conf         miredo.conf        OpenCL            python3.13        security          sudo.conf        vconsole.conf
apparmor.d        cron.yearly       fonts              hosts.deny         libao.conf         modprobe.d         openvpn           rc0.d             selinux          sudoers          vdpau_wrapper.cfg
apt               cryptsetup-initrfs  freedesktop        idnapt.conf       libaudit.conf      modules            openvpn           rc1.d             sensors.d         sudoers.d        vim
arp-scan          cryptsetup-nuke-password  fuse.conf          ifplugd           libblockdev        module-load.d      openvpn           rc2.d             sensors3.conf    sudo_logsrvd.conf  vmc
avahi             cryptedtab        gai.conf           ifplugd           libblockdev        modules            opt               rc3.d             services         sv               wgetrc
bash.bashrc       crypttab          geoclue           ImageMagick-7     libblockdev        modules            os-release        rc4.d             shadow           sys               wireless
bash_completion  dbus-1            geoclue           initscripts        libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          wpa_supplicant
bash_completion.d  dbus-1            geoclue           initscripts        libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
bindresvport.blacklist  dbus-1            geoclue           initscripts        libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
binfmt.d          default           glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
bluetooth         ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
ca-certificates  ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
chatscripts       ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
chromium          ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
chromium.d        ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
cifs-utils        ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
cloud             ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
colord            ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
console-setup     ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
crazklib          ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
credstore         ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
credstore.encrypted  ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
cron.d            ca-certificates  glib2              gnome-system-tools  libblockdev        mosquitto          pam.conf          rc5.d             shadow-          sysstat          xdg
```

Type to check for the list of users

cat /etc/passwd

Or just

cat passwd

Session Actions Edit View Help

└─\$ cat /etc/passwd

```
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:997:997:systemd Network Management:/:/usr/sbin/nologin
_dhcpd:x:995:995:DHCP Client Daemon:/var/lib/dhcpd:/sbin/nologin
systemd-timesync:x:989:989:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:988:988:System Message Bus:/nonexistent:/usr/sbin/nologin
tss:x:986:986:tss user for tpm2:/:/usr/sbin/nologin
strongswan:x:100:65534::/var/lib/strongswan:/usr/sbin/nologin
tcpdump:x:985:985:tcpdump:/nonexistent:/usr/sbin/nologin
sshd:x:984:65534:sshd user:/run/sshd:/usr/sbin/nologin
_rpc:x:101:65534::/run/rpcbind:/usr/sbin/nologin
statd:x:102:65534::/var/lib/nfs:/usr/sbin/nologin
dnsmasq:x:983:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:103:104:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
nm-openvpn:x:981:981:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
speech-dispatcher:x:980:980:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
nm-openconnect:x:979:979:NetworkManager OpenConnect plugin:/:/usr/sbin/nologin
pipewire:x:978:978:system user for pipewire:/nonexistent:/usr/sbin/nologin
saned:x:105:105::/var/lib/saned:/usr/sbin/nologin
lightdm:x:106:106:Light Display Manager:/var/lib/lightdm:/bin/false
polkitd:x:977:977:User for polkitd:/:/usr/sbin/nologin
rtkit:x:976:976:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:975:975:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
pcscd:x:974:974:PC/SC Smart Card Daemon:/:/usr/sbin/nologin
geoclue:x:973:973::/var/lib/geoclue:/usr/sbin/nologin
Debian-snmpp:x:107:108::/var/lib/snmpp:/bin/false
sshd:x:108:109::/nonexistent:/usr/sbin/nologin
stunnel:x:971:971:stunnel service system account:/var/run/stunnel:/usr/sbin/nologin
cups-pk-helper:x:109:999:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
redsocks:x:110:111::/var/run/redsocks:/usr/sbin/nologin
_gophish:x:111:113::/var/lib/gophish:/usr/sbin/nologin
iodine:x:112:65534::/run/iodine:/usr/sbin/nologin
miredo:x:113:65534::/var/run/miredo:/usr/sbin/nologin
redis:x:114:114::/var/lib/redis:/usr/sbin/nologin
postgres:x:115:115:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
mosquitto:x:970:970::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:116:116::/var/lib/inetsim:/usr/sbin/nologin
mysql:x:969:969:MariaDB Server:/nonexistent:/bin/false
_gvm:x:117:117::/var/lib/openvas:/usr/sbin/nologin
kali:x:1000:1000::/home/kali:/usr/bin/zsh
```

To add a user, type

Sudo adduser (followed by a name or title ex: adminuser)
I chose the user to have a name: mostafacyber.

```
(kali㉿kali)-[/etc]
$ sudo adduser mostafacyber
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
```

```
(kali㉿kali)-[/etc]
$ sudo adduser mostafacyber
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mostafacyber
Enter the new value, or press ENTER for the default
    Full Name []: mustapha
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] ☐
```

```
(kali㉿kali)-[/etc]
$ sudo adduser mostafacyber
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mostafacyber
Enter the new value, or press ENTER for the default
    Full Name []: mustapha
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y

(kali㉿kali)-[/etc]
$ cd ..

(kali㉿kali)-[/]
$
```

To switch to the new user by typing `sudo su mostafacyber`

```
(kali㉿kali)-[/etc]
$ sudo su mostafacyber
(mostafacyber㉿kali)-[/etc]
$
```

When the user `mostacyber` tried to install `wireshark`, he did not have access.

```
(kali㉿kali)-[/etc]
$ sudo su mostafacyber
(mostafacyber㉿kali)-[/etc]
$ sudo apt install wireshark
[sudo] password for mostafacyber:
mostafacyber is not in the sudoers file.
```

Now let's give `mostafacyber` the privilege to be in the `sudoers` group

To switch to the main user, just type: `exit`

```
(mostafacyber@kali)-[/etc]
$ exit
exit

(kali@kali)-[/etc]
$
```

Let's give mostafacyber the administrator privilege

Type:

Sudo usermod -a -G sudo mostafacyber

```
(kali@kali)-[/etc]
$ sudo usermod -a -G sudo mostafacyber
[sudo] password for kali:

(kali@kali)-[/etc]
$
```

We just made the user: **mostafacyber** an administrator.

How to remove a user account

Type:

sudo userdel (followed by the user you want to delete)

In our case: sudo userdel mostafacyber

And hit enter, and the account has been deleted.

